Explicit Nonlinear SU(2) Geometry on a Four-Dimensional Lattice Block

Quaternion bounds, incidence geometry, a local gauge slice, and a five-norm initializer

Mathematical research dossier

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Abstract

We give explicit finite-dimensional constants for the nonlinear plaquette-log map on the side-two four-dimensional cubical block. The argument represents SU(2) by unit quaternions, bounds the first three derivatives of a product logarithm by an absolutely convergent Banach-algebra series, and combines these bounds with the exact cubical incidence constants. On the logarithmic Coulomb slice the nonlinear curvature map is quantitatively bi-Lipschitz. A separate contraction theorem constructs the local Coulomb representative, modulo residual constant conjugation. Composing these geometric estimates with the exact heat-kernel derivative certificate produces all five local action-norm components required by a one-step retained/fiber marginalization. For $t \le 0.01$, four-link radius 0.003, and centered fiber radius 0.001, the scaled Hessian remainder is at most 0.747472, the centered-fiber transport coefficient is at most 0.266017, and the generated reference-base margin is at least 0.403230.

The same certificate isolates the next mathematical obstruction. At the upper heat time, the one-plaquette probability outside the certified curvature radius 0.006 exceeds 0.9999. Thus the finite local geometry is closed, while a probabilistically useful chart and a dependent bad-block polymer estimate remain necessary for a full regulated RG theorem.

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1 Result and mathematical role

The exact SU(2) heat-kernel action gives a gauge-invariant plaquette density, as developed in the lattice formulation of Menotti and Onofri [1]. A constructive block transformation requires more than the radial one-plaquette formula: the action must be transported through noncommuting link coordinates, gauge directions must be removed, and local derivative estimates must be aggregated over the block. These are the finite-dimensional operations treated here. Their organization follows the small-field side of gauge-covariant block RG constructions such as Balaban's [2, 3].

Let B be the cubical complex with vertices $\{0,1,2\}^4$. Write B_1 for its positively oriented links and B_2 for its positively oriented plaquettes. A link-log field is

$$A = (A_e)_{e\in B_1} \in \mathfrak{su}(2)^{B_1} \cong \mathbb{R}^{648}, \qquad U_e = \exp(A_e).$$

For $p = (x;\mu,\nu)$, $\mu<\nu$, define

$$U_p(A) = U_{x,\mu}U_{x+\hat{\mu},\nu}U^{-1}_{x+\hat{\nu},\mu}U^{-1}_{x,\nu}, \qquad X_p(A) = \log U_p(A). \tag{1}$$

The scalar incidence matrices are $d_0 : C^0(B) \to C^1(B)$ and $d_1 : C^1(B) \to C^2(B)$. They act diagonally on the three color components.

Theorem 1.1 (Calibrated nonlinear block certificate)**.** *Use heat generator $\frac{1}{2}\Delta_{\mathrm{SU}(2)}$ and let*

$$t_{\max} = 10^{-2}, \qquad r = 3\cdot 10^{-3}, \qquad \rho = 2r = 6\cdot 10^{-3}, \qquad r_F = 10^{-3}.$$

On the convex Coulomb chart

$$\mathcal{C}_r = \left\{ A \in \ker d_0^T : \|A|_{\partial p}\|_2 \le r \text{ for every } p \in B_2 \right\},$$

the following hold.

1. *The map $X = (X_p)_{p\in B_2}$ is injective and satisfies*

$$\|X(A) - X(B)\|_2 \ge 1.354892\, \|A - B\|_2, \qquad A, B \in \mathcal{C}_r.$$

   Consequently its inverse on its image is 0.738066*-Lipschitz.*

2. *Mean-zero vertex gauge parameters of norm at most $6\cdot 10^{-4}$ give a contraction with coefficient at most* 0.051367*. Every input link-log field of norm at most $2\cdot 10^{-4}$ has a unique Coulomb representative in that parameter ball, modulo constant conjugation, and the representative has norm at most $0.0026 < r$.*

3. *For the pure plaquette action*

$$S_t(A) = \sum_{p\in B_2} -\log p_t\big(U_p(A)\big), \qquad 0 < t \le t_{\max},$$

   the global scaled Hessian remainder relative to $\|d_1 A\|^2/(2t)$ is at most

$$\eta = 0.747472,$$

   and the global scaled third derivative is at most $\tau = 295.100$.

4. *In Hodge-whitened retained coordinates and an orthonormal conditional fiber coordinate, the five good-chart budgets may be chosen as*

$$\varepsilon_F = 0.747472, \qquad \varepsilon_{FY} = 0.528543, \qquad \varepsilon_Y = 0.373736,$$
$$\tau_F = 295.100, \qquad \tau_{FY} = 208.668.$$

With reference curvatures $\lambda_F = 2$, $\lambda_Y = 1$, these give

$$K_F \le 0.266017 < 1, \qquad \lambda_Y - \eta_{\mathrm{good}} \ge 0.403230 > 0.$$

These conclusions concern the declared local chart and its pure plaquette action. The density contributions of the link Haar measure and the gauge slice, together with the chart complement, remain separate analytic inputs.

The proof occupies Sections 2–5.

2 Quaternion product-log calculus

Identify SU(2) with the unit quaternions. For $a \in \mathbb{R}^3$ regarded as a pure imaginary quaternion,

$$\exp(a) = \left(\cos\|a\|, \frac{\sin\|a\|}{\|a\|}a\right).$$

Quaternion norm is multiplicative. The principal logarithm of a unit quaternion (q_0, q) before the antipode is

$$\log(q_0, q) = \frac{\operatorname{atan2}(\|q\|, q_0)}{\|q\|}\, q.$$

Lemma 2.1 (Banach logarithm derivatives)**.** *In a unital normed algebra with submultiplicative norm, let $z = 1 + h$ with $\|h\| \le \delta < 1$. Then*

$$\|D\log z\| \le \frac{1}{1-\delta}, \qquad \left\|D^2\log z\right\| \le \frac{1}{(1-\delta)^2}, \qquad \left\|D^3\log z\right\| \le \frac{2}{(1-\delta)^3}. \tag{2}$$

Proof. The series

$$\log(1+h) = \sum_{n\ge1} \frac{(-1)^{n+1}}{n} h^n$$

converges absolutely. The first derivative of h^n contains n ordered insertions, the second contains $n(n-1)$, and the third contains $n(n-1)(n-2)$. After division by n, summing the norm majorants gives

$$\sum_{n\ge1} \delta^{n-1} = \frac{1}{1-\delta}, \quad \sum_{n\ge2}(n-1)\delta^{n-2} = \frac{1}{(1-\delta)^2},$$

and

$$\sum_{n\ge3}(n-1)(n-2)\delta^{n-3} = \frac{2}{(1-\delta)^3}.$$

□

Lemma 2.2 (Product derivatives)**.** *Let*

$$P(a_1, \dots, a_m) = \exp(a_1) \cdots \exp(a_m)$$

with the direct-sum Euclidean norm on the inputs. For $k = 1, 2, 3$,

$$\left\| D^k P \right\| \le m^{k/2}. \tag{3}$$

Moreover, on the input ball of radius r, $\|P - 1\| \le \sqrt{m}\, r$.

Proof. The iterated Duhamel formula for $D^k \exp(a)$ is a sum over permutations of simplex integrals. Every exponential factor has quaternion norm one and the sum of the simplex volumes is one, so $\left\| D^k \exp(a)[u_1, \dots, u_k] \right\| \le \prod_j \|u_j\|$. Applying the product rule and then Cauchy–Schwarz to each differentiated input gives $m^{k/2}$. The final estimate follows by telescoping the product from the identity and using $\|\exp(a_i) - 1\| \le \|a_i\|$. □

Combining Lemmas 2.1 and 2.2 gives a closed formula. For $\delta = \sqrt{m}, r < 1$, set

$$M_1^{(m)}(r) = \frac{\sqrt{m}}{1 - \delta}, \tag{4}$$

$$M_2^{(m)}(r) = m \left(\frac{1}{(1-\delta)^2} + \frac{1}{1-\delta} \right), \tag{5}$$

$$M_3^{(m)}(r) = \frac{2m^{3/2}}{(1-\delta)^3} + \frac{3m^{3/2}}{(1-\delta)^2} + \frac{m^{3/2}}{1-\delta}. \tag{6}$$

Proposition 2.3 (Four-factor plaquette chart)**.** *For the oriented plaquette map* (1)*, if the joint norm of its four link inputs is at most r and $2r < 1$, then*

$$\|DX_p\| \le M_1, \qquad \left\| D^2 X_p \right\| \le M_2, \qquad \left\| D^3 X_p \right\| \le M_3,$$

where

$$M_1 = \frac{2}{1-2r},$$
$$M_2 = \frac{4}{(1-2r)^2} + \frac{4}{1-2r},$$
$$M_3 = \frac{16}{(1-2r)^3} + \frac{24}{(1-2r)^2} + \frac{8}{1-2r}.$$

If $L_p = d_1|_p$ is the oriented linear boundary sum, then

$$\|DX_p(A) - L_p\| \le M_2 r, \qquad \|X_p(A)\| \le 2r. \tag{7}$$

Proof. Signs in the two inverse links are absorbed into $a_i \mapsto -a_i$, an isometry. Equations (4)–(6) with $m = 4$ give the first three claims. At the origin $DX_p(0) = L_p$; integrating $D^2 X_p$ along the radial segment gives the derivative deviation. The last inequality is the triangle inequality for the bi-invariant geodesic distance on SU(2). □

3 Exact block incidence and quotient geometry

Direct cubical enumeration gives

$$|B_0| = 81, \qquad |B_1| = 216, \qquad |B_2| = 216, \qquad d_1 d_0 = 0.$$

Every plaquette has four boundary links. A link lies in at most six plaquettes, and a vertex meets at most eight links. Therefore, if E_p is the extraction of the four link variables around p,

$$\sum_{p \in B_2} \|E_p h\|^2 \le 6 \|h\|^2 . \tag{8}$$

The interval with two edges has vertex-Laplacian spectrum $\{0, 1, 3\}$. The Cartesian product rule gives the four-dimensional spectra by summing four interval eigenvalues. Consequently

$$\|d_0\| = \|d_1\| = \sqrt{12}, \qquad \|d_1 h\| \ge \sqrt{2}\, \|h\| \quad (h \in \ker d_0^T). \tag{9}$$

The ranks are

$$\operatorname{rank} d_0 = 80, \qquad \operatorname{rank} d_1 = 136.$$

The 24 displayed boundary-face fluxes have rank 17 on the physical quotient, leaving scalar fiber dimension $136 - 17 = 119$. Tensoring with $\mathfrak{su}(2)$ gives retained and fiber dimensions 51 and 357.

Theorem 3.1 (Nonlinear curvature coordinates)**.** *On the plaquettewise radius-r Coulomb chart, define*

$$\kappa(r) = \sqrt{6}, M_2(r) r.$$

Then

$$\|DX(A) - d_1\| \le \kappa(r). \tag{10}$$

If $\kappa(r) < \sqrt{2}$, the curvature map is injective on the chart and

$$\|X(A) - X(B)\| \ge (\sqrt{2} - \kappa(r)) \|A - B\| . \tag{11}$$

Proof. Apply (7) to every plaquette and sum squares. The incidence estimate (8) yields (10). For A, B in the convex chart,

$$X(A) - X(B) = d_1(A - B) + \int_0^1 \big(DX(B + s(A - B)) - d_1\big)(A - B)\, \mathrm{d}s.$$

The Hodge lower bound (9) and the triangle inequality give (11). □

For $r = 0.003$,

$$M_1 = 2.012073, \quad M_2 = 8.072581, \quad M_3 = 48.630394, \quad \kappa = 0.059322,$$

so $\sqrt{2} - \kappa = 1.354892$.

4 A quantitative local Coulomb slice

Let

$$V_0 = \left\{ \phi \in (\mathbb{R}^3)^{B_0} : \sum_{x \in B_0} \phi_x = 0 \right\}$$

be the nonconstant gauge-parameter space. Its orthogonal complement is the residual constant conjugation. For a link $e = (x, y)$ set

$$A_e^\phi = \log\left(\exp(\phi_x)\exp(A_e)\exp(-\phi_y)\right) \tag{12}$$

and define $F_A(\phi) = d_0^T A^\phi \in V_0$. At the origin,

$$D_\phi F_0(0) = -\Delta_0, \qquad \Delta_0 = d_0^T d_0|_{V_0}.$$

The product spectrum gives $\left\|\Delta_0^{-1}\right\| = 1$.

Theorem 4.1 (Local gauge contraction)**.** *Suppose $\|A\| \le a$ and seek $\phi \in V_0$ with $\|\phi\| \le s$. Put*

$$R = \sqrt{a^2 + 2s^2}, \qquad q = \sqrt{96}\, M_2^{(3)}(R)R.$$

If $\sqrt{3}R < 1$, $q < 1$, and

$$a \le (1 - q)s, \tag{13}$$

then $F_A(\phi) = 0$ has a unique solution in the radius-s ball of V_0. The transformed link field satisfies

$$\left\|A^\phi\right\| \le a + 4s. \tag{14}$$

Proof. Consider

$$T_A(\phi) = \phi + \Delta_0^{-1} F_A(\phi).$$

At zero, $A^0 = A$, so $\|T_A(0)\| = \left\|\Delta_0^{-1} d_0^T A\right\| \le a$. The three-factor product-log bound controls the derivative remainder on every edge by $M_2^{(3)}(R)R$. An increment in a vertex field is extracted at the two endpoints of each edge. Since the maximum vertex degree is eight, the global extraction norm is at most $\sqrt{8}$. Composing with $\left\|d_0^T\right\| = \sqrt{12}$ and $\left\|\Delta_0^{-1}\right\| = 1$ gives $\|DT_A\| \le q$. Condition (13) makes the radius-s ball invariant. Banach's fixed-point theorem gives existence and uniqueness. Finally, the geodesic triangle inequality on each edge and

$$\sum_{(x,y) \in B_1} (\|\phi_x\| + \|\phi_y\|)^2 \le 16 \|\phi\|^2$$

give (14). □

For $a = 0.0002$ and $s = 0.0006$, the theorem gives

$$R = 0.000871780, \quad q = 0.0513663, \quad (1 - q)s - a = 0.000369180, \quad a + 4s = 0.0026.$$

Thus the constructed Coulomb representative enters the radius-0.003 action chart.

5 Heat-kernel action and the five-norm initializer

Write the local heat-kernel action as

$$s_t(x) = \frac{\|x\|^2}{2t} + R_t(x) + c(t).$$

On $\|x\| \le \rho$, Stage 07 supplies

$$t\,\|DR_t\| \le g, \qquad t\left\|D^2R_t\right\| \le \varepsilon, \qquad t\left\|D^3R_t\right\| \le \tau_{\mathrm{HK}}. \tag{15}$$

For $t \le 0.01$ and $\rho = 0.006$, the explicit values are

$$g \le 2.00002 \cdot 10^{-5}, \qquad \varepsilon \le 0.00333336, \qquad \tau_{\mathrm{HK}} \le 3.20004 \cdot 10^{-5}.$$

Proposition 5.1 (Refined block aggregation)**.** *Let κ, M_1, M_2, and M_3 be the chart constants above. The scaled Hessian remainder of S_t relative to $\|d_1A\|^2/(2t)$ is bounded by*

$$\eta = \kappa(2\sqrt{12} + \kappa) + \varepsilon(\sqrt{12} + \kappa)^2 + 6(\rho + g)M_2. \tag{16}$$

The scaled third derivative is bounded by

$$\tau = 6\left[\tau_{\mathrm{HK}}M_1^3 + 3(1+\varepsilon)M_1M_2 + (\rho+g)M_3\right]. \tag{17}$$

Proof. For the principal quadratic part,

$$D^2\frac{1}{2}\|X\|^2 = DX^TDX + \left\langle X, D^2X\right\rangle.$$

The first difference from $d_1^Td_1$ is at most $\kappa(2\sqrt{12}+\kappa)$. The second is at most $6\rho M_2$ by the incidence estimate. The Hessian term from $R_t \circ X$ is bounded by $\varepsilon\,\|DX\|^2 \le \varepsilon(\sqrt{12}+\kappa)^2$, and its gradient term contributes at most $6gM_2$. This proves (16).

The third-order chain rule on one plaquette gives

$$\tau_{\mathrm{HK}}M_1^3 + 3(1+\varepsilon)M_1M_2 + (\rho+g)M_3.$$

For unit global directions h, k, ℓ,

$$\sum_p \|E_ph\|\,\|E_pk\|\,\|E_p\ell\| \le 6,$$

which proves (17). □

Substitution yields

$$\eta = 0.747471288, \qquad \tau = 295.099754.$$

5.1 Retained/fiber coordinates

Let $H = d_1^Td_1$ on the physical scalar quotient. Let C be the rank-17 retained coarse-flux map and N an orthonormal basis of $\ker C$. The conditional-mean subspace $H^{-1}\operatorname{ran}C^T$ is H-orthogonal to N. Choose M on that subspace so that

$$M^THM = I, \qquad M^THN = 0, \qquad N^THN \succeq 2I. \tag{18}$$

Since $H \succeq 2I$ on the quotient,

$$\|M\| \le \frac{1}{\sqrt{2}}, \qquad \|N\| = 1. \tag{19}$$

The same statements hold after tensoring with the three color components.

Writing $A = My + Nz$, the global Hessian and third-order bounds therefore give

$$\varepsilon_F = \eta, \quad \varepsilon_{FY} = \frac{\eta}{\sqrt{2}}, \quad \varepsilon_Y = \frac{\eta}{2}, \quad \tau_F = \tau, \quad \tau_{FY} = \frac{\tau}{\sqrt{2}}. \tag{20}$$

With $r_F = 0.001$, define

$$\begin{aligned} \mu &= 2 - \varepsilon_F, \\ h &= \varepsilon_{FY}/\mu, \\ \chi &= r_F(\tau_{FY} + \tau_F h), \\ K_F &= \chi/\mu, \\ \eta_{\mathrm{good}} &= \varepsilon_Y + \varepsilon_{FY}^2/\mu. \end{aligned}$$

The calibrated values are

$$\mu = 1.252529, \quad h = 0.421980, \quad K_F = 0.266017, \quad \eta_{\mathrm{good}} = 0.596770.$$

This proves the action-norm part of Theorem 1.1.

6 Probability overlap and the next inequality

The chart is analytically nonempty, but an RG proof also needs its complement to have a summable activity. The one-plaquette exceptional probability can be computed from the exact character expansion. At the calibrated endpoint,

$$\Pr_{t=0.01}\{\theta(U_p) \ge 0.006\} \in [0.9999423268758832,\, 0.9999423268759137]. \tag{21}$$

The omitted-character bound after character 200 is below $1.2 \cdot 10^{-86}$; a separate floating allowance is included in the displayed interval.

Equation (21) prevents a full-measure interpretation of the local certificate. The quantitative next target is an overlap window in parameters (t, r) satisfying simultaneously

$$\eta(t, r) < 2, \qquad K_F(t, r, r_F) < 1, \qquad \|\mathfrak{z}_{\mathrm{bad}}(t, r)\|_{\mathrm{poly}} < 1, \tag{22}$$

where the last expression is a dependent, gauge-invariant block-polymer norm, not a product of one-plaquette probabilities. An SU(2)-specific covariant derivative calculus can improve the first two inequalities by exploiting compact-group cancellations absent from the Banach-log majorant. The third requires shared-link dependence, Bianchi constraints, boundary holonomy, center flux, and topological sectors.

7 Verification and epistemic boundary

The accompanying implementation performs the following independent checks:

- exact cubical enumeration, ranks, incidences, and numerical spectral cross-checks;

- quaternion exponential/logarithm round trips and commuting plaquette identities;
- random noncommutative plaquette samples against the analytic Taylor remainder;
- gauge invariance of the plaquette class angle and numerical convergence of the Coulomb fixed-point map;
- cross-check of the five local budgets against the separate Stage 06 one-step calculator;
- hostile rejection of cut-locus, excessive-radius, lost-margin, and false mass-gap promotions.

The theorem proves a local finite-block statement, classified at claim level C2. The following bridges remain open:

Claim	Status	Missing object
Quaternion, incidence, local inverse, and local Coulomb certificates	P	Proved in this note
Five-component pure-plaquette good-chart norm	P	Proved in this note
Executable finite-block constants and hostile tests	N	Reproducible code and JSON
Link-Haar and gauge-slice determinant in the same norm	O	Density derivative estimates
Dependent chart-complement polymer activity	O	Probability-overlap theorem
Cutoff- and volume-uniform repeated RG	O	Quasilocal interaction recursion
Reflection-positive OS continuum Yang–Mills theory	O	Construction and theory identity
Positive Hamiltonian mass gap	O	Basin entry and spectral transfer

No finite-block coercivity constant is a physical mass. The result advances the constructive program by making the first nonlinear representation bridge explicit and by locating its first quantitative failure at the interface between local chart size and exceptional-set probability.

References

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